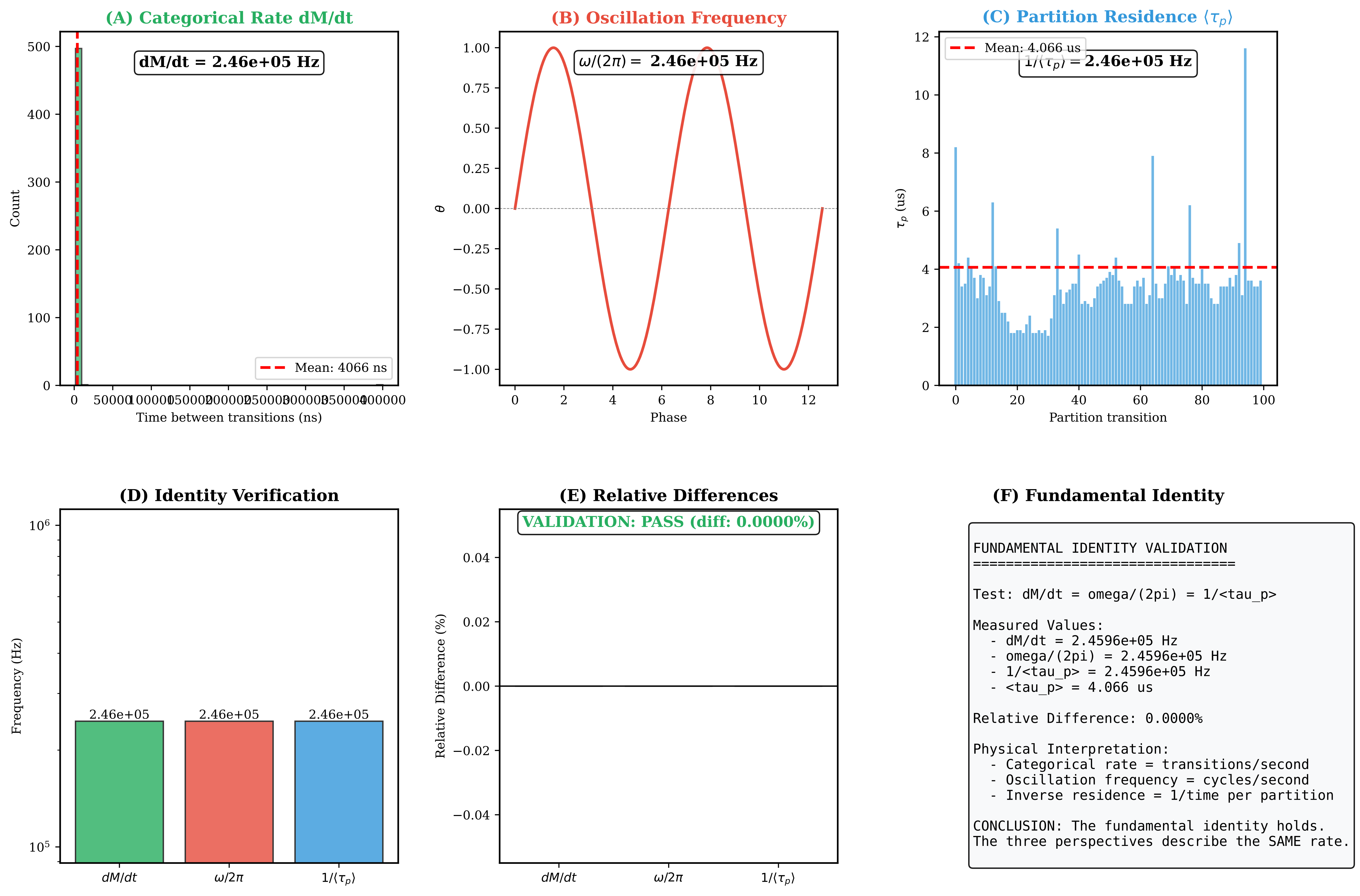

(A) Categorical Rate dM/dt
dM/dt = 2.46e+05 Hz
Mean: 4066 ns
Count
Time between transitions (ns)
(B) Oscillation Frequency
ω/(2π) = 2.46e+05 Hz
θ
Phase
(C) Partition Residence ⟨τ_p⟩
Mean: 4.066 us
1/⟨τ_p⟩ = 2.46e+05 Hz
τ_p (us)
Partition transition
(D) Identity Verification
Frequency (Hz)
2.46e+05
2.46e+05
2.46e+05
dM/dt
ω/2π
1/⟨τ_p⟩
(E) Relative Differences
VALIDATION: PASS (diff: 0.0000%)
Relative Difference (%)
dM/dt
ω/2π
1/⟨τ_p⟩
(F) Fundamental Identity
FUNDAMENTAL IDENTITY VALIDATION
================================
Test: dM/dt = omega/(2pi) = 1/<tau_p>
Measured Values:
- dM/dt = 2.4596e+05 Hz
- omega/(2pi) = 2.4596e+05 Hz
- 1/<tau_p> = 2.4596e+05 Hz
- <tau_p> = 4.066 us
Relative Difference: 0.0000%
Physical Interpretation:
- Categorical rate = transitions/second
- Oscillation frequency = cycles/second
- Inverse residence = 1/time per partition
CONCLUSION: The fundamental identity holds.
The three perspectives describe the SAME rate.